

ANGELA ZHANG

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WORK EXPERIENCE

Gameplay Software Engineer II
EA Maxis Redwood Shores, CA Current Role

- Simulation engineer working on Project Rene (unreleased), architecting fundamental gameplay systems in C++ and visual scripting languages

Gameplay Engineering Intern
EA Maxis Remote Summer '22

- Led the development of a data-driven Unity-based prototype in C# for a new gameplay mechanic to answer crucial LOD design questions
- Formulated key algorithms working extensively with navigation, scheduling, and procedural generation

Software Engineering Intern
Microsoft Xbox Remote Summer '21 & '22

- Designed and implemented a metadata class in C# to support a large-scale data migration from DynamoDB to CosmosDB for PlayFab, a backend platform hosting over 2.5 billion player accounts for 5000 games
- Created an IP Geolocation Kubernetes service to provide player location information to game developers
- Automated weekly updates of database files from a Cosmos Virtual Cluster using Github Actions to ensure data accuracy
- Worked extensively with Terraform, FluxCD and Azure Resources to integrate service into existing PlayFab Kubernetes clusters and create a shareable package for cross-team access

Algorithmic Design Intern
SeisWave Houston, TX Summer '19

- Developed model-building algorithms in C to automate the denoising correction and analysis of seismic data, greatly improving accuracy and efficiency over industry-standard methods

PROJECTS

C0 VSCode Language Support
VSCode Extension Nov '19 - Jan '20

- Designed and released a course-specific VSCode extension in TypeScript used by hundreds of CMU students each semester with over 10k downloads
- Improved the efficiency and understanding of student coding by providing language support features including auto-completion, syntax highlighting, and go-to

Bigger, Badder, Buggier
Web Game - UI + Gameplay Jan '21 - May '21

- Worked with a team of 20 on a Unity game that captured the spirit of CMU Buggy Racing during the remote semester
- Implemented core gameplay mechanics for an engaging racing experience, such as AI systems for CPU buggies, rubber-banding speeds and power-ups

EDUCATION

Masters in Entertainment Technology
Carnegie Mellon University | Entertainment Technology Center
Graduated 2023

B.S. in Computer Science
Minor in Game Design
Carnegie Mellon University
Graduated 2022

TECHNICAL SKILLS

C++ C# C Java Python

JavaScript TypeScript React Node.js

Unreal Unity Kubernetes Docker

Git Perforce

PUBLICATIONS

Kasarda, J.P., Zhang, A., Tong, H. *et al.* (2025). **The Oomplet dataset toolkit as a flexible and extensible system for large-scale, multi-category image generation.** *Sci Rep* 15, 9287. doi:10.1038/s41598-025-93036-y

Kasarda, J. P., Bahuguna, J., Zhang, A., *et al.* (2026). **BitterBuster: a virtual world for understanding naturalistic patch foraging.** Preprint.

BitterBuster + Oomplet Dataset Toolkit
HCI Research Software Sep '22 - Dec '22

- Designed and built a game in collaboration with the CMU Psychology Department to evaluate the effectiveness of Human-AI collaboration on foraging and identification tasks
- Created a robust open-source Python toolset capable of generating 9.1 million unique visual stimuli across ten controllable feature dimensions that can be used for studies on perceptual learning and classification

Summer of Longmian
Unreal Game Demo Jun '22 - Aug '22

- Worked with a small indie team on a demo reminiscing about spending a summer in the Chinese countryside; garnered strong interest on Chinese social media with over 400k views
- Lead programming development in C++ and Blueprint Visual Scripting, building UI, minigame, and quest systems